

ActInf GuestStream 047.1 ~ "Predicting & Reflecting Framework for Dual Process Theory" Bellini-Leite

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hello and welcome this is active guest stream 47.1 on July 10 2023 we're here with Nick bird and Samuel Bellini lychee and we're going to have a presentation followed by a discussion so thank you both for joining and Sam thanks again for this presentation off to you let's see uh was Nick gonna say something before I started oh I can go ahead okay okay uh so some introduction first I'm a professor Minister State University I'm a psychologist philosopher mainly working in cognitive science I'm going to talk about what I did in my DHC thesis and also some renewed interests I had on this based on some developments in large large language models which happened this year uh most notably chain of thoughts and tree of thoughts those additions to Varnish language model made me want to come back to this again and talk to this talk about this so how do you know about human reasoning which is the subject the main subject underlying this topic well we had Empirical research from the 60s starting with Peter Watson who was challenging the notion that we have a logic logical models for reasoning or some general understanding of understanding of logic when we're adults and he crafted tasks which were hard to to solve but they had actually simple fundamentals to it uh which shows which we're taking to scene taken to show people's irrationality and something like that and also a huge tradition that's called Logistics and biases and also started in the 60s and the 70s mainly by continent and almost first king which had similar tasks and went beyond and they even won the the Nobel Prize for this work yeah I I had here in the slides evolutionary psychology and language and pragmatics which were also people in the 90s working with this sort of research but I'm gonna talk about dual process Theory uh in the way they are formulated by Evans and stanovic which uh mainly summarizes the results of Watson and and [Music] so I'm going to talk about how do process Theory came to be and then some problems that it has that I try to address so one of the first first reasoning tasks in in this Trends or literature was the Watson selection test which goes like this please indicate each cards can determine if the following rule is false if the card has a vowel on one side then has an even

number on the other side um this sounds like a simple exercise so the card a clearly is relevant because it has a problem on one side which is if it doesn't have an even number on the other then the statement is false and everyone can find this neatly of most people but the problem comes with the the card seven which is not so intuitive people actually choose the second the second cartoon there which has an even number on one side but having that even number doesn't help at all to determine if the statement is false what does help is which is an odd number if it has a vowel on the other side then the statement could be so this was one of the tasks that was famous for University children this one they're famous task as well it's called the consumption 2000 the issue that people derive and solve this sense it is uh because it goes like this it is a 31 years old single outspoken and very bright she majored in philosophy as a student she was deeply concerned with issues of discrimination and social justice and also participated in anti-nuclear demonstrations please make the following statements by their probability is important for the most probable and eight for these problems uh what's interesting here is the disease statement the last statement Linda is a bank teller and it is an active is active in the feminist movement because there's also there are also these statements without one another without the conjunction which is active in the famous movement and then there is a bank account because of the description people think there's a feminist is very likely given the description but they think being a bank teller is not likely the issue is that people uh usually marks Linda is a bank teller and is a feminist as more likely than being the main seller the problem here is that the conjunction of the two necessarily is less likely so being a famous plus being something else or being a bank Target plus being something else is automatically less likely so those are the sort of tasks that the reasoning literature used to show some sort of human irrationality uh one issue with this is that very interesting to show that humans are irrational it's more interesting to show how we how humans reason and also these sort of studies didn't show how people actually solved the the Departments they were actually focused on how people made mistakes uh one thing that's interesting about dual process theory is that explains both the mistakes and the solutions to to the answers so in 2002 Carmen and Frederick formulated this cognitive reflection task which is another one of these tasks but it's a lot more simpler and elegant and it really shows indicates at least how dual process theory is a good solution to a good explanation of the evidence in general yeah the Western selection test used logic in the Linda task use probability and here we just have basic arithmetics so basic math you don't have to to worry about which which answer is truly correct or which rational Norms we need to follow it's just basic management it's there's nothing to discuss here right so

the test goes like this a bat involved costs one in ten in total you might know this as it's a famous sense even out of the reasoning literature the bat costs one more than well how much does the ball cost so people when people read this they usually say 10 cents intuitively but if you take a little time to write write it down you can see that it's actually five cents so uh the bat costs one and five cents and the ball costs five cents together and it costs one insane uh the other two have a similar structure the point here is that sometimes we look at these tasks and we have an intuitive answer which is wrong and the way we get out of these these issues is that you stop and think a little bit and review our answers with some sort of step-by-step reasoning so writing it down something like that uh what's interesting here is like if you look at two plus two it's almost impossible not to think four it's just it's almost like you're perceiving the answer is four just by looking at it but if you have some sort of more complex calculation to do obviously just by looking at it you won't have a solution so this basic uh intuition is clearly shown in the cognitive reflection task and what's more interesting about it is that the cognitive reflection test actually predicts the performance of various emissions and biases tasks so if someone does well on the cognitive reflection test they're more likely to go well on any other reasoning task uh and if they go bad on the cognitive transaction test that you'd like to go back on other resume tests as well this is also to me this is also an indication that this dual process structure of intuitive reasoning versus slow reasoning reflective reasoning likely explains the rest of the tests as well not only do you sue but most of the others so it's a good explanation of the literature okay so what does the theory say uh decides besides explaining the the reasoning literature so in 2008 Jonathan Evans tried to make a formulation of it that would include and unified views from different different domains such as social cognition neurosciences uh other people in psychologists we had which had similar theories and it seems like there's something in in uh important here to unified but Evans wasn't able to and so there's a formulation specific for reasoning and we don't know if it applies so much well to to other areas of technician well one common way to describe dual process theory is by making these tables of features so we say oh type 1 processes have these sorts of features on the left in type 2 processes like this other type of features on the right uh this table here was one I made by joining some of the features of Evans and Stanovich and also of Kahneman and then with some questions I raised in the thesis I elaborated it this way but it's very similar to the events and standard HTML formulation one problem we have with this tables formulation of the developed process theory is that usually people agree that in one thinking instance when you're solving a problem you're not going to have all of those features

together so we don't know for sure when some feature will be present not for instance so clear that type 1 processes necessarily have to be under conscious and type 2 processes necessarily have to be conscious uh I think that comes a lot because of some conceptual issues in the theory but one solution they have found was to have at least some defining features but different authors have different defining features so here I have some of the defining features of the the main authors which is type 1 processes use less working memory uh once had two processes loads strongly on working memory type one processes part autonomous in this sense that they don't need to wait for higher processing to respond if they have an answer they go for it and using the couple's representations mean that means that type 2 systems can reason Beyond what's what's in the present environment you can think about things that are not there uh type one processing is faster in comparison R2 is lower in comparison type 1 processing is less effortful well two is more effortful in the sense of psychological effort effort like people reports having a hard time doing things for instance in the cognitive reflection test people that solved the test correctly usually rate it is harder because they had to use the type 2 processing to solve it okay but Samuels which is a philosopher a dual process theory in other subjects he says okay even if these features proposal is good you see we still need a mechanism to explain why these features are linked together what what is the mechanism that explains why type 1 processing features go together and what is the mechanism that explains why type two processing goes together so we need some Beyond saying that they're different you need to explain what are these mechanisms that make them differ and I actually had Samuels online PhD committee and I'm not sure if he was convinced with my solution so my PhD was a solution to this problem I'm not sure how how he was convinced of it but he approved the the PHD so that's good enough I guess yeah the way I so the goal was to solve the unity problem is to explain the mechanisms that that explain type 1 processing and explain the mechanisms to explain type 2 and I started to see to search in the literature first how people were conceiving of the general Frameworks that were behind these two types of reasoning and what that was not actually related to this literature of dual process Theory but was actually famous in cognitive science was the modularity of mind uh in the 80s it was my first for General folder and he said that uh we had two types of gross and cities the modular process mostly related to input and perception then we had Central processes that was mostly related to higher reasoning so you can Trace some sort of resemblance there to do process Theory and the way he went about this is that he said these perceptual models the modular processes are information encapsulating so Vision only works with vision and language only

works with language space when it goes to space and so these were domain-specific knowledge systems that did not communicate with each other and when they needed contact sensitivity they needed Central processes so these Central processes he would argue were isotropic and finance which means that any type of information could be used in a solution so we can relate I don't know physics with Biology and form a new solution so there's Knowledge from multiple domains working here and that that which they would also need some sort of context sensitivity sensitivity which was a problem for photo if you didn't think we would solve this context sensitivity problem at all and he had good reasons for that in classic AI we had a lot of issues with that I mean you still do have issues with the consensus context sensitivity problem but not uh the sort of issues we had in the 80s when they were debating this same problem of classical AI so this problem was a problem with how to represent a changing environment in symbolic representation uh so yeah it was a problem because classical AI attempted to give knowledge to robots so artificial intelligence and statements about the world and they were far intractable hard to compute there are so many knowledge events to search in so many knowledge to try to use the the robots couldn't do anything relevant uh they also couldn't representate them represent a change so the changing environment was terrible for them didn't work with this symbolic representation it didn't work with serial reasoning or anything like of the sorts so this was a huge problem for plastic cognitive scientists and since photo was a believer in cognitive science in the sense of classical computational theory of mind or language of thought Theory she said that he had found for his first law of the non-existence of cognitive science this law states that the more processes are Global the less cognitive science will be able to understand them so he thought only these encapsulated processes could be studied and if we needed this sort of global and contact senses context sensitive cyber processing then contest science just can do it because he had in mind going to science in the sense of the classical AI but it also seems not only folder it seems like the literature abandoned classical cognitive science in various ways which were strange because if you take the work of Alan new and Herbert Simon it's strange to think that oh they were completely wrong there's nothing nothing useful that they they say that could be applied today all the evidence of the theories they made but just completely wrong that's that's a real weird way of thinking and well if you take for instance our theories today of the human Beijing brain they don't consider this at all so it's just like they they don't get a different strand theories but it also it always seems to me that there must be at least something wrong about uh something right about this shouldn't be totally abandoned there must be a place for these

results in this explanation of the Mind one thing that was in arts with the way forwarder described this dual process Theory and also sometimes stand of age when he talks about dual process Theory it seems like it is forgetting this fact here is the fact that we have a limited capacity when we use conscious processing or and this is one of the best evidence in Psychology we have so far we've had behaviorists evidence for reinforcement and enforcement learning and that stuff and then we had cognitive psychology in the 60 and this is the best evidence that cognitive psychology has had for a long time so this is also something we should not simply ignore so we have in short-term memory the knowledge that when we store some information in short-term memory we usually can only keep track of about seven items sometimes even less in selective attention there was this task where there were there was a video of people playing basketball the subjects had to count how many passes they were making and suddenly a gorilla passed by on the scene and people didn't notice the gorilla because they were so concentrated on the which which is some evidence of selective attention that is when you're concentrated on the task you actually don't even see other interestingly we have also a competition for limited resources when doing dual tasks or if I'm talking on the phone and writing at the same time I'm probably not going to do a very good writing because we're competing for a limited resources and it's important to note that if you're walking in in riding and walking and talking so those are different tasks they won't compete so we had some competition we have more more competition when these limited resources are disputed also in working memory so if I say the number seven nine three two four one and I ask you to repeat them backwards and you're not looking at the slides you're going to have a hard time with that because of the limited capacity is worth memory when it's clear that in other areas or thinking of our our base using a lot more information in certain items to to make the scenes happen and perception and to make us walk and so on so that's on the side of the type two constraints on the side of type one reasoning so common are already has some intuitions similar to the ones I described so he said performance earlier's day days the research that diversity and I conducted was Guided by the idea that intuitive judgment Supply position between the automatic corporations of perception and the deliberate operations of reasoning Carmen and Frederick claimed that intuitive thinking is perception like that intuitive predictions is an operation of system one further that the boundary between perception and judgment is fuzzy and permeable the perception of a stranger as medicine is inseparable from the prediction of future harm so you can see that this is very likely and it's very similar sorry to the intuitions that we're speaking on when we're talking about prediction processing so kind of saying when you look

at a face of someone and he's mad uh you have a prediction that you're in trouble right so this is a sort of judgment that seems for perception since front reception also the concept the contact sensitivity is pretty much salt in the predictive processing and if you take on the AI literature that's also happened for instance in the paper attention is all you need they have some uh tips for how to deal with context sensitivity which disorder didn't dream of so this is something that Clark said in 2013. the best overall fit between driving signal and expectations will often be found by in effect inferring noise in the driving signal and does recognizing a stimulus as for example the letter m saying the context of the word mother even though the same Bears was presented out of context or in most other contexts would have been a better fit with the letter n a unit normally responsive to the letter M might under search circumstances be successfully driven by and and like stimulus so what this Clark is trying to say here that since we have we represents using probability then if the contest says we should go left to go left if the context says we should go right we do right because the representation is fluid and it accepts changes really quickly so that's something that's changed from the predictive processing architecture but one issue I think that we might see in the predictive processing architecture is the lack of compositionality so symbolic representations exhibit compositionality meaning that complex representations are built by combining simpler elements according to rules or syntax the meaning of a complex representation is derived from the meanings of its constituent Parts the way they are combined this property allows for the generation of new and meaningful Expressions by manipulating simple instructions so solar in 88 execution argue that even as connectionless models succeed they would likely have some sort of stimulation of the language of thoughts that exhibits conversationality in order to work like we do and you see that it seemed like it's the case with the results we have this year uh so considering those constraints and the initial view of order and the problem I was trying to solve which is the unity problem from Blue process d a major shift I had on the PHD thesis and then I wrote a chapter on this book which handbook and found in rationality with Keith so we explained here in this book The the chapter in this this shift the power structure which is I mean it's not clear that you know a dual process theory is theories would adopt folder but it's still very similar uh the way the received field of dual process Theory Works which is well type one is a number of dumb heuristics blind modules and animal intelligence and then type 2 is the real contextual and complex reasoning the real human intelligence so I tried to invert that power structure by by saying that the contextual predictive processing is is a type one reason which which is giving us the basics to reason on so our reasoning already comes with

context we don't need to reason the context with type 2 uh we all use type 2 reasoning to fix some minor issues in predictive processing when we use heuristic searching because it's limited uh we only use it when there's a lot of prediction error in predictions aren't working that well and then we call for this heuristic search to handle whatever is missing to see if we can find a new solution that wasn't captured so in this case the powerful system is a predictive processing and not the the type 2 usage search right so that was the general idea for this shift that now I'm gonna say some hypothesis that stems from this shift so T1 processes deal with content encoded in the form of probability density functions which is how predictive processing uses to represent the world which means there is no symbol and no definite content those values means the standard deviation Muslims by previous movements in the previous World contingencies manipulating prior information biases the distribution into one or another Direction closer to or further from a certain value so people here in the active Insurance Institute who know this by heart I'm not changing anything here I'm just using what they say basically these functions are not stored in the memory bank but distributed from the responsible brain regions over to the external organs and body parts to neural connections the values in the distribution do not represent objects directly in discreetly they refer to distance X aspects of the input when perceptual systems are dealing with such objects this is in line with the T1 processes being easily biased when working with references to similar properties like similar numbers objects Rhymes or pet names very often and the incorrect value is picked from a distribution an example that I have from this is when usually I have seen people confuse their youngest child with their dog the dark knee with the youngest child and when someone has a another new child then the the youngest the new youngest child is often confused with the dog because they are I think somewhat represented in singular distributions significant to the distribution this is also the language claimers of embodied proposals that the world is not represented in synthesis T1 processes are super personal and their predictions are made by the same systems which we process perception a clear example is that a judgment about a facial expression is related to the FFA which is a region we bring that processes faces and this is also related to that judgment that continent judgments about their face the idea is that perception is not passive but already comes with predictions and when in problem solving such prediction is precisely the type 1 answer so if we look back at the the reasoning tasks all those type 1 tasks are likely stemming from predictions I don't know what to claim that you want processes are purely perceptual using consensus superlative only that such prediction is staying for your perceptual processes so what is perception and currencies I think the word

you just using different words for this is the problem so just not not getting into that convinced example of judgments of negative spacious Expressions how this is expected this new process Theory like uh it's expected of dual process Theory the type one reason Works under predictive processing yeah also in line with the claims of your body cognition that there is no sharp link between perception and reasoning on the other hand C2 processing works like a classical machine for reasoning such as general Problem Solver of Newman Simon part of this classical machine only made sense in the brain if it's if it exists in a wider setup it's predictive processing Network generating type 1 responses liking news physical symbol system when facing reasoning problem T2 processing opens a problem space containing an expression the designates the initial problem and an expression designates the solution which was positioned was productive produced by probabilistic prediction having the initial expression in the predictive expression in the problem space C2 processing then uses its smooth generators to attend to reduce differences between them and sometimes find different solutions and such bad so let me just explain this a little bit better when neuro was working in the 60s there they they knew they couldn't search all possible space of a proper space because it wouldn't work so they have this have this search which works like the detectives Let's ignore the problems which are likely wrong and just look at the spaces that might be right right uh what this predicting in reflection framework does for a music search is that the risk of search only works on it works already with the prediction that was estimated from type one processing so it's like we only start reasoning and making searches based on the prior predictions we already had right and we all we only go further with the search and the predictions aren't working and the main point here is that the recent search isn't searching a random space or even a pre-programmed space is searching a space that was left out from the predictions that started in the type 1 processing so that's how it doesn't go to the frame problem anymore because a large part of the contextual issue was also already solved by predictive processes Okay so that was the General hypothesis I presented to attempt to solve the unity problem which is the problem of how type 1 features go together and how tattoo features were getting and now I'm going to try to argue that this is likely the case since that I can prove this is the case but there are good reasons for us to think that type 1 processes are use predictive processing and that types of processes do not use predictive processing rather they need some some kind of symbolic heuristic search one thing that's that stands out of is is a good explanation for the difference between implicit representation in explicit representation if you look at the literature in psychology you will see that it's very ambiguous in this these definitions like people use

implicit and explicit representations as the same thing as unconscious conscious or fleeting graphical or the same thing as automatic in control and so when it goes to dual process Theory it's not very clear but they're saying in which case but they use this expression implicit and explicit a lot how I think this model solves this is by saying that implicit representations are probability density functions while explicit representations are symbolic representations classical symbolic symbolic representations uh Clark in his book sometimes mentions that we can have single Peak distribution distributions but the problem with having single Peak distributions is that well if you only have one Peak on the solutions function then it's nice it doesn't have all these these features of the probabilistic representation it's more likely similar to a symbolic representation in fact that's having a single Peak distribution might be the way that probability representation Ally symbolic representation these have different and important features so the probability representation is continuous it's uncertain and ambiguous and that's why it's able to be sensitive to context a new example the mother example of choir it can vary on retrieval we can remember something and be different in the second time uh it's tied to priors it works with statistical relations yeah it's it's called in and depending on which value is most probable at the time it has a different outcome right so I argue that explicit representations are likely discrete and there are likely symbolic in nature they're unambiguous this is what allows us to disappear in the first place they're stored reliably if you have a reliable definition of it you likely remember it the same way in the next sign uh it's arbitrary in the sense that it's not really coupled to the stinger in the steam line in the sense that the of perception usually is the statistical relations are related to aspects of the world uh it's using compositionality and it's immutable in the sense that you can change the you can't change it it's fixed this value doesn't change so that's the implicit versus explicit feature you can have also explanations for ultimaticity contrast to working memory so automaticity concerns overlearned skills and Overland skills are understood as skills that have become predictable in this framework so you usually use this to explain like driving riding bicycle riding um so when you're driving if you don't know how to drive you need certain statements like uh some statements about how you how you shoot steer right you need to have those in mind but then once you learn how to drive it's like your body uh already knows what it's doing and you don't even need to think about this so this is the classical distinction in Psychology but I argue that the predictive processing Theory makes a new enlightening to this so it's like our body is predicting what we're doing so it's automatic so it's an explanation for ultimate CC but say that a dog certainly uh suddenly runs from his car you obviously obviously don't have a model for

that so you're gonna need to call in working memory to solve some of those issues and so you don't hit the dog so everything works under prediction unless the models very unreliable and then working memory is called in to solve some further issues an interesting hypothesis I had on the PHD thesis which I never explored and I actually don't even think anyone read this part but it seems to me to be a very interesting hypothesis in relation to free energy and active infants so that's why I'm bringing it here in the active influence Institute that as we know the brain attempts to minimize free energy by getting predictions right thus higher need of type 2 processes are related to Higher free energy interviewing information in the sense when the predictions are working clearly then you don't need to to have effort but when there are a lot of prediction error then you need effort because you'll call in the working memory to do a risk surge so to minimize a great amount better amount of free energy will take more time in work this is more effort flow than having predictions ready the minimize free energy as quickly as possible before when probabilities fail the system needs to service game possibilities it searches for other possible solutions by means of heuristic surge the reset search would be related to more information and time because it does not have probable Solutions right instead it needs to investigate its state space almost from scratch we say almost because it is illustrating hence it also will have tricks to get the correct solution faster unlike Brute Force search which would investigate the state space from beginning to end we believe reports of effort will be related to executing more risk searches there's reports of effort by subjects would seem to be based on cognitive information and physical constraints of reality so that's why I think this is an interesting part of the hypothesis is because we have this psychological measure of effort but if this is true then we actually have a physical measure of efforts like if people are working hard on a problem that means there's a lot of free energy going on and they are having to use this heuristic search to fix some issues that the model isn't able to switch by itself foreign yeah so that's that's the effort part so now going for the working name report what's interesting here is that um in working memory which memory it's like it's it's expected of it that's that it works like a classical computer working memory is widely researched Topic in technology but in predictive processing it's readily rarely mentioned and I actually searched the predictive process in the books it's Ctrl F it's working memory and people rarely use this this topic it's not done by predictive processing it has nothing to do which produce this processing and it's actually an issue for predictive processing because well if you're gonna eliminate classical symbolic processing all together then we need a predictive processing explanation Authority memory and so far I haven't found one and I don't think there will be

one but this also could be just ignorance of mine maybe there's some paper there which I hadn't found okay wrong but in any way is very aligned with what touring was thinking when he was making the the connection between machines and mines so if this is a sentence of Alan Turing explaining his computer right the turing machine and if you change the word computer to working memory it clearly works it's almost like it's talking about the same thing when of course if we change the word here computer to predictive processing it's clearly you're not talking about the same thing so the behavior of the computer or working memory at any moment is determined by the symbols which it's observing in his state of mind that moment we may suppose that there is a bound B to the number of symbols of squares which the computer can observe at one moment if he wishes to observe more humans use successive observations we will also suppose that the number of states of mind which can be taken into account is finite the reason for this are the same characters as those which restrict the numbers of symbols if we admitted an Infinity of states of mind some of them would be arbitrarily closed and would be confused again the Restriction is not one which is seriously effective computation since the use of more complicated states of mind can be resorted to by writing more symbols on the same so this is very similar to what we think about when we're talking about working there not only because we were inspired but because of the evidence that concerned working memory is very similar to what suspected of this sort of machine now finally we have speed so predictive processing has basically various strategies to make the the processing B is faster as fast as possible while that's not true for symbolic classical AI so yeah Clark says cheap fast won't exploring action rather than the pursuit of Truth optionality deductive inference is now the key organizing principle and he's talking about predictive processing uh the predictive processor is always taking certain bets about what the current state that we were implies losing accuracy in compensation for Speed we also have a predictive coding in the the more strict sense so when predictive quality we mean specifically the property of this system to consider from the world only stimulant which result in Greater prediction error so there's also a filter there which allows for Speed and and focusing on predictive prediction relevance system that only domestic agents who quickly decide forces of action and select the moves to amongst the possible sources so predictive processing is tailored to be fast necessarily well that's not the case for symbolic Ai and so that's that's the explanation for the difference in speed so beyond that the two processes need to figure out the solutions online so it's different if you have a prior prediction that would say to you what the answer is from having to search a space from from scratch right there's also another issue that

the biological brains certainly were not built for a Serial optimistic search and so the Republicans say that the models that the absolute speed of a process is a property or Excellence of its implementation in sensitively brain does not have a symbolic processor implemented if we do use some sort of heuristic search it's like a um a different adaptation it's not it's not what the brain is is used to the insertion problem spaces is slower than having a problem outcome ready with the stimulates okay so that was the work I did in um my PhD thesis and those one those were one of the main reasons why I think these framework is good for explaining the differences in features of dual and I actually wrote a prediction that I I just I recently learned that I wrote this because I I forgot I wrote this and then I went back to to the thesis and then I said well I actually said this would happen and and it kind of did so that was one of the reasons why I I renewed my my interest in my PhD thesis so what I said was predictive processing system would be subject to bias from lack of compositionality such as mistakes and creativity failures and noticing necessary character from rules and so on precisely the type of mistake type 1 processing encouraging so at that point we didn't have large language models affecting uh attention is all you need paper came out in the same year and and then I noticed that this could be seen as a prediction of the type of issues that large language models are facing that's we out reliability part-time keeping the order of units or steps in complex reasoning or maths trade our time letting go of priors and to my what really motivated me to to start working at this again was that they solved this precisely by using dual process theory in T2 agents that are calling so what they're doing now to solve these reliability issues is adding a heuristic search to the generative models and and it's it's having awesome results this is just this year it's in 2023 the more they develop these type 2 agents to answer these reliability problems of large language models the more the large language models are getting good in that stuff so this is this is a paper I got this is the images from the paper tree of thoughts right this one which is one of the last ones probably came out of them about two months ago something like that the way they're they're doing this implementation is they're taking the uh the the um results the outputs of the large language model and they're doing further reasoning on it and then feeding the the large language model with the result of the further reasoning the further reason they're they're doing is very similar to the classic symbolic AI so here they explained there's a simple input output prompting here which is simply inputting a prompt and the large language model will give you an output and as we know these these outputs sometimes are very biased there's hallucination on it they're not reliable you know all of that stuff and last year they figured out that through chain of from chain of fat

protein we can also already increase the reliability of this these models so you give an input and just as we do the better prompting you know when we go to channel and say I'll do this better fix this issue solve this step by step in some of the the commands that are good for chapter and Chain of Thought protein automated some of that some of those reasoning steps and then large language models were working much better enough to this so this was last year and then last year this started a new trend of research to to make these agents coupled to a large language model outputs the large language models to see how they can get better um the self-consistency one you give the input and then there are various Solutions the self-consistent the sales consistency forces the large language model to try different solutions and then you get a majority vote so if you have like five similar answers in in some different paths then this one Speaks Over the one which is less popular and the tree of thoughts which I thought was most similar to the the proposals that I had in in psychology is the one which opens up the problem space is it's actually pretty much what what I just explained in the places I want to open up a problem space and they start to search possible solutions which are better than the original and they feed that back into the uh predictive processing oh sorry the alternative model the large language model and then this this one is uh usually it's currently one of the best ones yeah uh so here I made a general diagram of how this would work it's not psychology nor AI it's like what's what's similar in the two so we have a genitive model here using probability density functions this looks more like a traditional qualitative connection is model than a predictive processing Point that's just it's just the diagram it's not that good I'll talk a little bit about that so yeah we generate an answer here which is to the problem here and we can get a linguistic output and this linguistic output then goes for an heuristic search using symbolic representations yeah so if you get a better answer he repeats it back to the generative model and so on basically the knowledge part is the genitive model and the illusive search is simply using more steps to see to prop it back it's more like a prompting scheme than a new knowledge scheme which is both true for the Dual process theory model I created and the the heuristic search that's that's happening here on the t2 agents for LMS a better way to do this and I think we'll start those implementing this is that the input layers are trying to predict the thoughts right it's a bit more like predictive process in the other diagram and so they're trying to predict the thoughts and by opening up this and searching this problem space during these these steps we already have new new prompts here for the general model so if you take predictive processing seriously then this does happen in the brain is more likely like this I do think that the tree of thoughts paper does say that the intermediary process isn't deferred in the

searching on the elements finally I thought maybe we could suggest something from psychology to to these models to keep keep growing working differently based on what we know from psychology so some things that are not implemented yet that is actually functioning have these updating abilities in intelligent for getting which is real it's related to Insight problem solving is when you you forget something but that it means like you forget to Prior right you're using a different prior So you you're able to yeah let go of biases right there's also work on thinking dispositions which could be relevant in this position additions or sentences like this release should always be revised in response to new information of our evidence so this would these will be like imperative linguistic knowledge which would have to be added on I don't know where but likely would help at work there's a lot of thinking disposition thinking dispositions that are relevant to our successful reasoning and this this is just one example yeah the literature and there was a lot more examples that could be relevant uh we have in creativity research we have this generative phase which is similar to what the genital models are doing but we do not have the exploratory processes going on which is common in creativity research literature but it's what we do with for instance me Journey when we're doing better prompts uh so maybe using these exploratory uh from the creativity research we can also automate the exploratory part of the the creativity processes that these image generators are getting to and not just the generative part and finally I think the embodiments we don't have robots or anything like that that do this sort of reasoning so AI on the computer and also stuff from active inference which was not considered slightly related to embodiment in the sense of navigating a world and predicting the world that stuff is far from happening in the traditional elements we have so yeah there's a lot of stuff said maybe I hope you guys you guys have some comments as I said uh I haven't been able to talk about this to anyone I do think I have some at least some maybe some relevant stuff that I've published learned since I'm not famous and I don't have people to talk to it's mostly gone unnoticed so thanks for the space here thank you awesome presentation all right Nick it'd be awesome to hear your introduction and then take it wherever you'd like to go sure so first thanks uh Samuel for interesting talk and I've been really pleased uh to find your research I guess maybe I'm supposed to say something about uh who I am where I am so I'm Nick bird I'm at the Stevens Institute of Technology in the New York City metropolitan area uh in a department that's kind of interdisciplinary so we have like philosophy and quantitative social science but also like people doing like music and visual arts and all sorts of other things um I tend to do more of the philosophy and quantitative social science stuff um so I'm kind of also interested

in human reasoning and so much much of the people I'm reading inciting and drawing on in my research are a lot of the people that you saw in the opening slides that we saw you know people like wayson and um Kahneman diversky and mercy and sperber um and a lot of the other people uh that were cited um I think Samuel does a really good job to incorporate the kind of like shoulders of giants that we're standing on in terms of the philosophy of mind right the the voters and the positions and the these people um and one thing that I think is uh you know on everybody's mind these days is these large language models so I'm really glad to see how this this dual process theory that was that emerged from research uh like economy and diverskies on these these problems like uh you know the conjunction fallacy problems or the wayson selection card task or the uh conjunction fallacy tasks how that has been applied to models like large language models and what we can learn from that and then I think the coolest thing about the pre-print is how we could kind of Port back and forth the learnings from both the computer science and the the cognitive science to improve one another right so I think that one of the neat things about this preprint that um Samuel was giving us at the end is that you know we're kind of taking what we think is a model of reflective that we like you know we represent um certain parts of a problem and and maybe reason more carefully and effortlessly about them in certain situations in ways that might help us and how can we help these large language models do the same because it does seem like large language models function at least the ones that we interact with online mostly they seem to function mostly as like a system one or type 1 process they're just kind of like quickly generating lots of uh text or imagery or something like that but they're not necessarily like reflecting on it and the ways that we would think that a uh you know a human is capable of um and they might not even be capable of like representing things in the same way that we do um yeah so I thought that was a really interesting idea this using Chain of Thought and tree of thought models to kind of create a reflective level or a reflective system within the models it seems just like a really valuable idea and just a really great synthesis of research and multiple disciplines something that I think few people are actually very good at is like incorporating some of the best insights from multiple Fields were often pretty siled in Academia so I just think this is great work and I hope more people will appreciate and pay attention to it and build on it um one of the things that stood out to me in this presentation more so than when I was like reading this 2023 pre-print analytic reasoning for large language models by Dr Bellini leiche um that thing that stood out to me is this idea that like billin Elijah and Frankish sort of Switched which of these two types of reasoning processes are supposed to be context dependent or context sensitive or however

you would want to word them and I started thinking well I wonder if there's a sense in which both types of responses or both types of reasoning are somewhat context dependent and I'll just say what I mean and then maybe um Samuel you can kind of say what you think I should think or or clarify the view of the predictive and reflective framework or something so the thought I was having was well in a familiar context these intuitive predictions these type 1 processes are gut response so to speak those are going to be pretty useful because they're well trained where we have a lot of experience that we're drawing on so like our gut response our first response is often quite good it's in these like less familiar contexts or maybe similarly familiar but like way higher Stakes or something but it's in these other contexts where we might think oh maybe I should slow down and make sure I've double checked whatever my initial impulse is before I just accept it because there's a lot writing on this or like I'm just not used to this this type of problem so I need to you know I need to slow down and so there's a sense in which what what context is doing is not just um showing up in one or the other type of reasoning but it's sort of like determining which type of reasoning might be best and at the moment um but I'm wondering like is that just totally compatible with what you're imagining or is that like somehow a deviation from the framework the the predictive and reflective framework yeah thanks for the thoughts there and compliments and um thanks for the question as well the reason why I think uh contexts needs always some sort of type 1 help is because of the constraints we have learned from symbolic AI so if I'm saying that type 2 reasoning have the same constraints then it cannot be contextual because it doesn't know it doesn't work like that we know that if the sales in contacts it feels bad but of course we do need to like when we have a novel situation we all often solve this by by context right so how could this be and I think it's by the um interconnection of the power of the tooth system so so we always start with a prediction which in this case would be mistaken and then we start we have to search for novel instruments and in this search uh we make the contextual uh comparisons it can't be that the system two is making the comparison on its own I think I think is uh more likely that when it's a case of context that stems from a novel situation and then the two systems have to figure out together likely in the sense that I was winning out here in the end these predictions that go over the thoughts the most likely are the ones that will solve these context issues in the novel problems yeah I I'd argue something like that that's somewhat helpful so then maybe what I'm thinking is something along what the lines you were saying at the end you were saying how there's still kind of an opportunity to understand things like executive function uh in this framework and I think that's maybe part of what I'm wondering about and I'm wondering so I

guess I'm wondering if yeah if there's more to be said within this predictive and reflecting framework about how each of these two types of processes get selected or like what might help the system that specific point you just remember I just remember something that that's related so yeah clearly our working memory or type 2 reasoning likely does more than that it does this but likely more more than that so there's there's likely also a belief bank something like the taking dispositions some sort of knowledge bank which is related to type 2 processing um so Keith Frankish does this distinction between flat out beliefs and I I'm not remembering the terms but he has a type 1 belief in type two belief type 1 believes is that but if you have but you're not fully confident of it and the the type 2 believes is the beliefs you have when you you have a say a very firm political position and you stated with obvious war is that you for sure you believe that so there's likely to be some sort of different belief structure as well which I don't talk about at all in thesis or the work so I'm not saying obviously that this is a complete model of reasoning uh and obviously we need more to to figure out how reasoning works so yeah yeah okay that's that's also helpful um I'm wondering if Daniel did you want to weigh in well I think it's a very interesting question about which of those modes are engaged or how they're balanced through how does this connect to what's known as the context window in today's Transformer type large language models so how do you map the computational attributes of large language models today like architecturally or their Ram or CPU usage how do you map that onto for example human cognitive processes do you think that's useful or there any insights there well I'm I'm mostly making a relation between predictive processing in cognitive models but I know that's I mean large language models are not entirely the same as what's active influence the same I'm aware of that but there is some similarity specifically in regards to regenerative model parts so I would I'm not sure I'm not a AI specialist to to be sure which details of regenerative model could be true of the human brain and I'm most likely betting that uh the the people who study cognitive models in the brain like the friskins you know have figured out something like what our generative models do one other thought or Nick do you have it you want to add there um just thought of some different ways whether people have connected it explicitly to the literature on working memory in predictive processing I I think as you pointed out it's definitely a link that is not highlighted it's one that we recently heard um from uh professors Walker and uh Manriquez and Friston in the recent livestream 53 that was on like cognitive paleoanthropology looking at human working memory but some ways that people have incorporated working memory is like a nested model that carries context at a deeper time but that is not necessarily like an actual mechanism that

gives rise to why type 1 and type 2 are the way they are um so I thought that was a very provocative direction to go to move past the descriptive and then to look at the underlying generative model of why these outcomes are the way they are even though these cognitive phenomena are causes of other things happening they are also caused by some influenceable or contextual or variable aspects and when we lose that context dependence of the cognitive systems then they're totally lifted and disembodied don't really help by putting the Primacy on that predictive processing element I think it opens the door to connecting those areas Beyond just mentioning the terms in proximity but to really support different epistemologies from predictive processing as a model or approach yeah can you go on that idea again I'm not sure what you want me to comment can you summarize that again and make it a question sure what do you think the epistemological consequences of taking predictive processing the way that you have approached it here versus alternative or prior approaches to cognitive Sciences okay but what do you mean by systemological Advance with this Theory um how it influences our our understanding of how we know or seek or practice or do or decide perfect okay so let me let me think about this yeah I think one of the best uh what this helps with most is explaining dual process Theory because dual process theory is in bad shape in terms of Concepts uh in terms of theory and formulation people have no idea what these these systems refer to in the mind brain uh we often have you know different intuitions into this and basically basically what I most think this work is relevant for is for saving dual process Theory from criticism which has been happening in Psychology so people are like it seems like you're talking about the left and right brain right something like that something that we don't know for sure what these two systems are right so this is a good way forward I think uh for for dual process Theory and as a sad little processity does have some value in itself uh so yeah saving the process Theory I think it's a good direction for so this is this model but also if we if we think about advantages of it would be something like for instance it could have helped us had the idea of using chain of thoughts or or tree of thoughts had anyone read this before so it can help us make these these sources of predictions that we shouldn't have some sort of Serial reasoning coupled to the genitive models so yeah I think you can help us think on further on on AI if we uh AI like like like Nicholas saying to like incorporating more stuff in Psychology than we know but it's certainly missing here but here we have a direction of what we should incorporate and so on yeah but yeah I'm also unaware of the working memory research from predictive processing you mentioned and if you want to go back to that and explain me some more of that I'd be happy to hear thanks yeah sure so not that this is the most

effective way to implement a memory system but at least that this provides uh analytical method for measuring or describing them if one we're going to have the five digit number you can imagine a a nested model where the lowest level is the ones place and it's nested within a decision tree of tens places and hundreds and so on and then there's some seek or access policy or strategy that helps speak the number in reverse and some cognitive or computational limitation is just how well that cognitive agent can perform on that test again that doesn't mean that's the mechanism that's being used but that would be a way to use a nested generative model to encode like multiple levels of spatial or temporal variability but context becomes an issue because you're basically just expanding the the possibility space looking for sparser and sparse or associations so if you don't have a good compositionality or really well-definable well-articulated causal process then you're just building these all by all models that you're going to be searching through in the dark yeah one final and a comment on that is that well it's not clear that the best way to reason will be the way humans reason right so we might eventually have a different way of implementing a working memory on an llm which would be more effective than ours there's also a lot to the other side there's also the point that it seems that the limited capacity we have on working memory maybe it's not just a lack of power maybe it's necessary so we kind of like don't lose our minds in the sense that we have a limited space to search and then we're able to search the disc limited space and if it was too uh if you have if we had too much space to search because we had a super super working memory then we wouldn't be able to finish the tasks in time so maybe this is like a physical good uh barrier that's something that helps and it's interesting that what I find is that okay the act the AI may find a lot a lot of different better ways to implement worth memory but it's interesting that they implemented a virtual memory or a search agents which was very very like the way I described in the thesis that was the well they found amazing as you were saying there's there's very ways Implement they worked in memory but they did it in a very similar similar way in the sense of searching the space after the generative model has offered a solution right so I don't know maybe that's a a good a good uh indication that this is on the right track but it also may just simply be a simple way they found to to solve the problem and it was just by coincidence similar to this so yeah maybe the correct way or the way the brain does may also be different we don't know right all we have it was the The Limited evidence I I offered a bit more obviously and uh some constraints we know on AI and information processing and you have to figure it out and I think that this implementation stuff we're asking is mostly mostly due to the AI people to find out or the the math paperwork assistance to figure out Nick do you

want to comment or a question or I'll ask one from the chat yeah so I just have a kind of a bigger picture question that kind of gets us into the realm of these chat Bots um right so you in this pre-print and um in this talk have like given us a variety of different tasks in the heuristics and biases literature the the conjunction fallacy to uh task the Linda problem some people call it um or these cognitive reflection test questions and um you know there's a there's like a paper in the fall that showed that a preprint uh in the fall that last fall that show 2022 that showed GPT 3 and 3.5 and even the earlier versions like showed a lot of the human-like faulty intuitions and I can basically fell for the lure pretty often sometimes like most of the time it was falling for the lure but then as soon as chat or gpt4 uh was was available to be studied and used on these models it was like performing basically near perfect and when it did get the incorrect response it wasn't necessarily the lure it was just like some other General type of error it was making like misinterpreting the question altogether or something and it seems like the predict the predicting predictive and reflecting framework could have at least two different ways of explaining this like one is like uh what happened is they changed the AIS like type 1 thinking or whatever so that like it could uh respond intuitively to all uh to all these but intuitively and correctly so it didn't need to reflect or it could be that like they somehow created this like reflective type system with either like you know a Chain of Thought or a tree of thought or some other type of system that helps it actually engage in reflection on these these tasks or whatever the analog of reflection would be for a chat bot um and so I'm wondering like if you have thoughts on this like as you've been thinking about this or talking to anyone about this like which how do you think they did this obviously we have to speculate because a lot of the data is proprietary but I'm just curious to get your thoughts yeah that's an awesome question yeah I actually forgot to mention this on the presentation that the uh the the biases they make on the recent test last year were exactly the ones the the humans made so if we if we uh said it's um dual process real it would give exactly our mistakes that's that's so amazing I'm not certain why that happened but it did how they fixed this so yeah I do think that they trained the model on these tasks because people were able to like uh get similar similar issues by changing the questions a little bit so I think they like cheated on that in the sense that they trained on unknown problems but that's not true of new problems if you should give it some different formulation and feed it to I don't know 3.5 new version it's still it's still failing but uh the people on open AI they they are aware of this chain of thoughts research they are aware of this tree of thought research but I don't think they want to Depend and rely on these external agents so they recently uh released a report where they are trying to reinforce

intermediate steps of reasoning on gpc4 so they try to simulate internally what the the chain of thoughts reasoning is doing externally by reinforcing the intermediate intermediate steps of reasoning so that it stops feeling composition helping so on okay yeah I'll add a thought on that I was reminded by Daniel Dennett framework or model In Darwin's dangerous idea of cranes and Sky Hooks and so some building is built and then the question is like was it built with cranes or with top-down hung Sky hooks that just sort of descend from nowhere to hold everything up while it's all being built and it's like well no it's with cranes and then if you need a big crane you can use a smaller crane to assemble a crane so that's kind of a bottom-up constructive metaphor whereas the sky hook is like the top down compositionality and that's when you get to certain level of sophistication for a cognitive system it can make the blueprint or the plan like maybe what we would associate with the expected free energy calculation not just the variational free energy calculation so at some point when a plan can be made it requires a strategy between type 1 and type 2 switching and probably other switches too um and so that crane approach you need a lot of compute to do a bottom up so even though it's kind of weird to think about it's almost like the current large models are very bottom up because they're very bottom up from syntax and they don't have um what you refer to as imperative linguistic knowledge which can be normative as well as a heuristic for knowledge and that's potentially a small but it contains wisdom that's explicit and practices and um stances or or dispositions as you described and so it's kind of like it's not a denial of embodiments to also clarify what this sky hook ability is and it's type 2 like so I think I guess what do you think about that or how do you connect this to anything in that area yeah but if you do the summary in the question again because although I didn't track some of what you mean I'm not sure which ones made to comment on that do you think that the uh landscape of large and small models would be different if people took on board some of the the features you described like imperative linguistic knowledge seriously into construction of models so let me just see if this is something you're asking so they're they're doing small models now for phones and open source models performance and they have the issue of not being able to scale up right because if you scale up too much then you can't run the form uh so you're asking if we can Implement something like uh tree of thoughts to to have these small smaller models yes yes are you sure are you just saying okay go for that that's an example of what it would mean to re-understand like where do we need an API call to a cloud Center or where could some context be locally computed with much more of a type one what are some type 2 problems that we might be able to type one or way out of what are some type 1 areas where things are

not working and a type 2 um description might be advantageous yeah yeah I think they're they're getting on to that and they're they're being able to make these smaller smaller models precisely because of these sorts of um discoveries they made on chain of thoughts right there's also stuff about being able to train on specific data I read one it's called textbooks are all you need the the training smaller smaller models on the textbooks and they're doing better so yeah there's there are differences in the training oh training smaller models and different data but there's also this the possibility of making them more reliable based on what I'm calling T2 agents or train of thoughts stuff like that and again we don't know if that's all we can do externally but I would argue that we we would need necessarily some sort of external mechanism coupled to any large language model for the best results because of the differences in representations mainly because of this right unless the large language model somehow uh makes a symbolic representation emerge from its processes it's likely that it has done so but unless it's it does so and it uses that reliably because they may even have generated some sort of symbolic process internally but doesn't always use that or doesn't know when to use that so yeah most likely we're going to need something external to the language model and um yeah this this is some Clues to to how to implement it to implement that and also I think in our brain it's kind of external not only in the sense of our minds but also in the brain since it's a recent step in evolution uh we don't see I don't know okay and blizzards reasoning in type 2 manner right uh we do see chimpanzees originally in type 2 manners but not as much as we do so it's very human-like reasoning uh it's just that going back to the context problem animals do solve contextual problems that that's why they can they can live right so the issue is not actually the context there it's more like uh based on reasoning capacity in that boost on missing capacity may be related to the prefrontal cortex as we know and so of course the prefrontal cortex is also a network it's not a Serial turing machine but if uh it's it's there's somewhere it's implementing a Serial machine or a symbolic classic AI machine or symbolic reasoning or something like that most likely it would be the prefrontal cortex which is kind of like a distinct addition an addition to the brain and maybe something related to language is also which is also it's uh distinct a new uh addition to the brain so yeah and also I'm very convinced by the solar Indonesian argument in 3dh okay you're gonna need symbolic processing anyway right either the model is gonna generate decimal processing or you're going to have to feed it for any because or else you're going to have the whole destination or mistakes and reasoning because of the nature of because of the nature of distributed representations right it's it's uncertain in ambiguous by Nature that's why I'm not entirely

convinced that only probabilistic representations would be enough to solve continuation issues
awesome Nick do you have any other questions or areas you want to mention or ask I think
um I think those were my main questions um I don't know if there's anything in the chat that
you wanted to touch on before we go I know we're maybe over I'll go with one from the chat
and then we can have any closing thoughts so glia maximilisk glia maximalist asks are you
familiar with visual Pathways in the human brain specifically the dorsal wear and ventral what
Pathways do you think this maps onto the two systems you describe for problem solving okay
yeah so like I said into in 2008 Evans tried to accomplish that unifying vision of due process
Theory I'm not very convinced by that I mean he he also said well we can't do this as of yet
but I I do think it's possible as I said on the presentation I just don't know how to do it uh how
to have a um convincing Theory of Everything of dual process theory in the brain uh this it
does go on to other regions like you have automatic motor responses and controlled motor
responses and and like you said there's a different different Pathways in reasoning uh in
figuring out stuff and the uh visual field so yeah maybe the what would be related to
something and would wear to another most likely the what would be would be tapping on to
type 2 processing but I'm not sure how the where it would be tapping on that but yeah as I'm
trying to make the point if we extend it to the whole brain it we lose what we we've learned
we're not sure about that applies well Nick any penultimate last words uh you're muted oh wait
unmute and then continue sorry about that um thanks uh thanks to both of you uh Dr Bellini
latte and uh Daniel for coordinating this and I'm really just glad to have found uh the research
and to discuss it and connect and uh looking forward to further conversations to be honest
awesome Sam any closing thoughts yeah I'm just sad I couldn't go much uh beyond what I
said in the presentation yeah most of the stuff you you guys asked uh I'm not sure I have good
answers for that basically what I've done is all I know and I'm not sure classical uh on the
future with this and maybe someone in the artist audience will do so I hope that that helps
great I hope so too all right until next time thank you bye